

Trend Following, Leveraged Re-Entry, and Volatility Decay

(working title — headline to be finalised)

A reproducible, beginner-friendly study of whether **switching between daily-leveraged and ordinary S&P 500 exposure**, timed by a single 200-day moving average (the daily equivalent of Faber's 10-month rule), can improve long-term returns — tested honestly on total-return data going back as far as the record allows.

Disclaimer. Educational research only, not investment advice. Leveraged ETFs are high-risk and can lose most of their value. Past performance — real or simulated — does not predict the future.

Note on framing: this version follows the analysis step by step and leads with the methodology and evidence. The abstract/headline below is provisional.

How this study is structured

We deliberately test **only one trend signal — the 200-day moving average** — exactly as Faber popularised it (his rule is the 10-month SMA on monthly data; 200 days is its daily twin). The argument is built in six steps:

1. **Buy & hold vs the Faber moving-average rule.** Establish that the simple trend rule improves *risk-adjusted* returns.
2. **Leverage returns.** Define daily leverage and show what *constant* 1.5×/ 2.5×/3× exposure does to the index and versus the trend rule.
3. **Volatility decay — and timing.** Explain the decay, then show that buying leverage *at major market lows* (GFC, COVID, 2022, 2025) pays enormously — so leverage can help *if you time it*.
4. **How much leverage is "too much"?** Map, against trend and volatility, the leverage at which volatility decay exactly cancels the return (flat), and the leverage that merely ties 1×.
5. **The switching strategy.** Use the 200-day MA to switch leverage on and off. Test the intuitive *"buy leverage low"* version (leverage below the MA), then **check whether doing it the other way round is better** (leverage above the MA). Full metrics tables and equity/drawdown plots.
6. **Robustness & verdict.** Monte Carlo optimal leverage, real leveraged ETFs, costs, sub-periods, limitations.

Data

Everything is total return (dividends reinvested), from standard public sources, cached locally for reproducibility.

Series	Span	Role
Shiller monthly S&P price + dividends	1871–present	reconstruct monthly total return for the Faber replication
Reconstructed daily total return ([^] GSPC price + Shiller dividend yield)	1928–1988	early daily history
Real [^] SP500TR (true daily total return)	1988–2026	spliced on
[^] IRX 13-week T-bill (+ 3.5% constant before 1960)	cash / financing	

Reconstruction check. Over 1988–2026 the reconstructed daily total return tracks the *real* [^]SP500TR with **0.50%/yr** tracking error, **0.9996** correlation (CAGR 11.41% vs 11.47%). The full-period (1928+) max drawdown of –83.9% matches the 1929–32 crash, consistent with Faber's –83.66%. The long daily series is therefore safe to use, and we apply **true daily leverage** to it.

Step 1 — Buy & hold vs the Faber moving-average rule

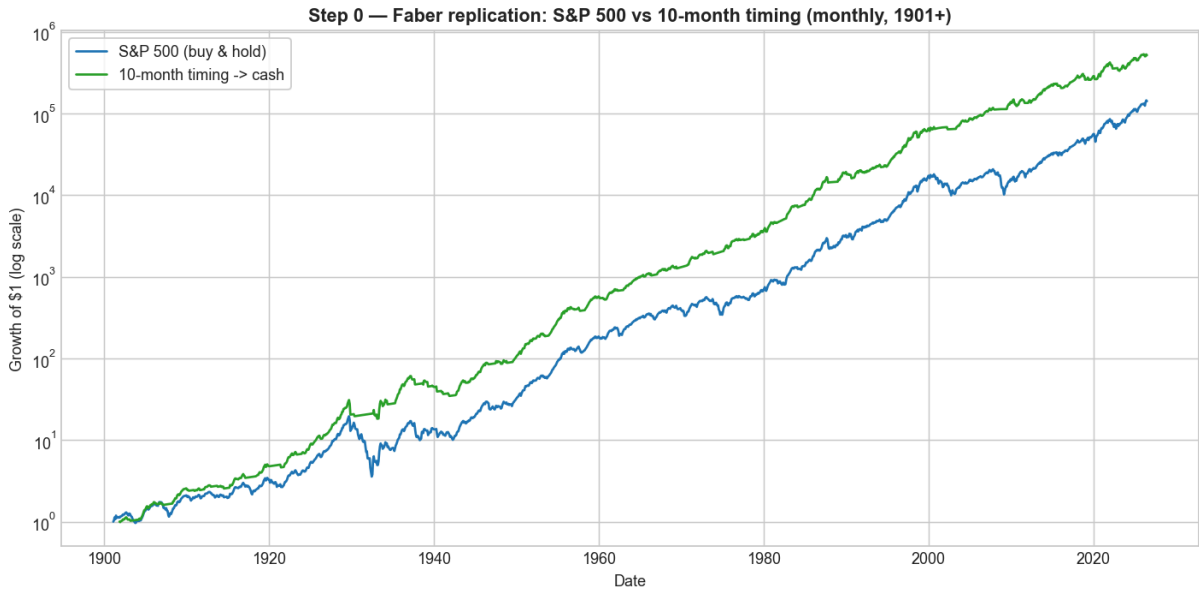
The rule (Faber): at each month-end, if the index is above its 10-month SMA, hold the S&P 500; otherwise move to cash (T-bills). We replicate it exactly, then use the daily 200-day twin for the rest of the study. The signal is always lagged (we trade on yesterday's close), so there is no look-ahead.

Monthly replication, 1901–2026 (Faber's setup):

Metric	S&P 500 buy & hold	10-month timing → cash
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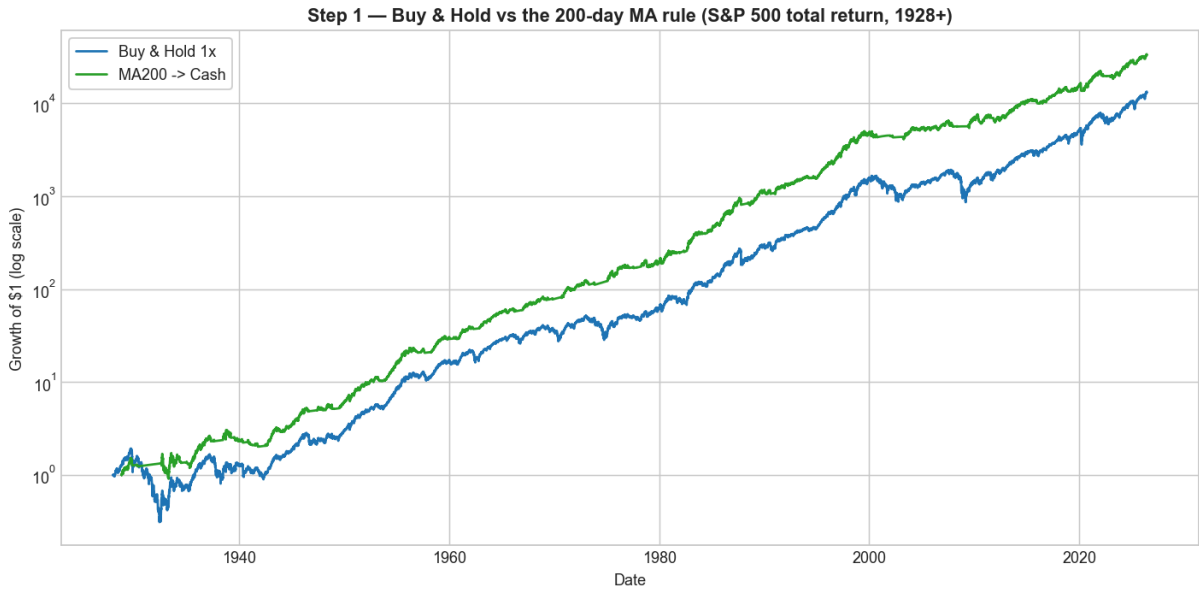
CAGR	9.95%	11.24%
Volatility	15.4%	10.8%
Sharpe	0.44	0.69
Max drawdown	-81.8%	-43.0%

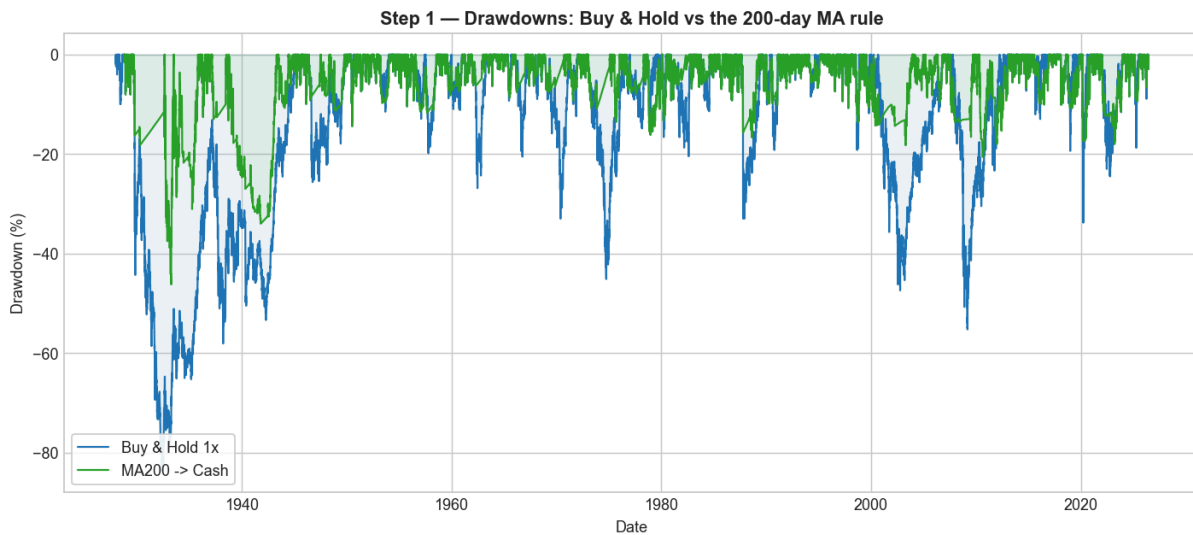
Faber's published figures are -83.66% → -42.24% for the same drawdowns — we land within a point, confirming the replication is faithful.



Daily 200-day version, 1928–2026:

Metric	Buy & Hold 1×	MA200 → Cash
CAGR	10.14%	11.29%
Volatility	18.9%	12.6%
Sharpe	0.40	0.60
Sortino	0.56	0.84
Max drawdown	-83.9%	-46.2%
Calmar	0.12	0.24





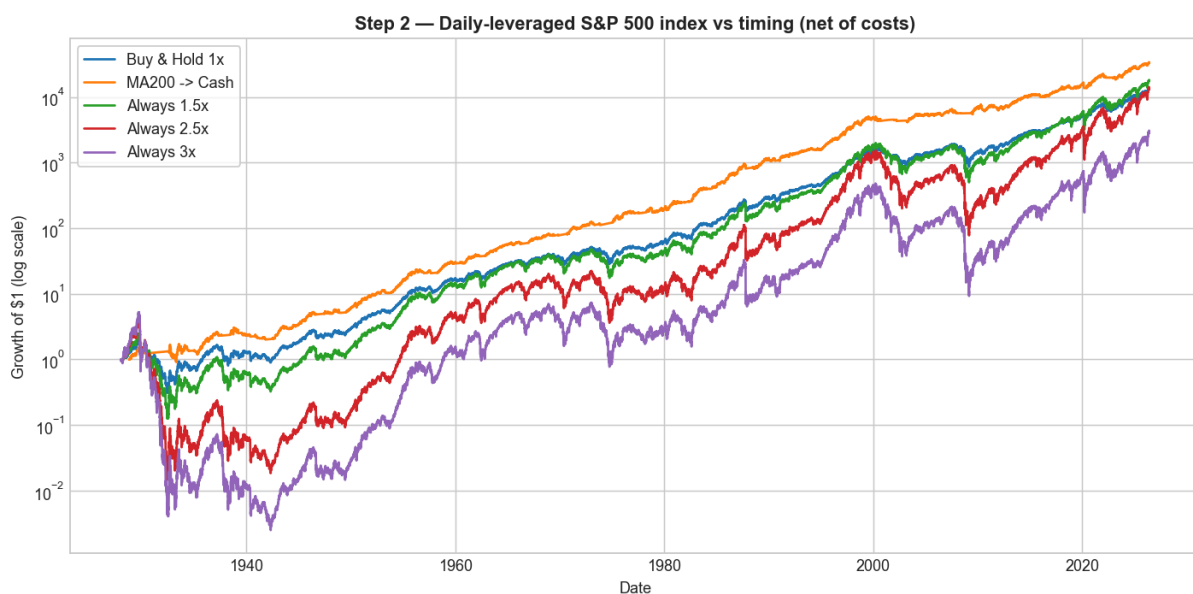
Takeaway: the trend rule keeps essentially all of the return while cutting volatility by a third and *halving* the worst drawdown — Sharpe, Sortino, and Calmar all jump. **The 200-day MA clearly adds risk-adjusted value.** That is the foundation everything else builds on.

Step 2 — Leverage returns

What "daily leverage" means. A daily $L\times$ fund multiplies each *day's* return by L and rebalances: $r_{lev}[t] = L \cdot r_{index}[t]$ (before fees/financing). It is *not* $L\times$ the multi-day return — the difference, after compounding, is volatility decay (Step 3). Borrowed money costs the broker call / financing rate, and leveraged ETFs charge $\sim 0.9\%/yr$; we include both.

Holding *constant* daily leverage on the index over the full history (net of costs):

	CAGR
1× (buy & hold)	10.14%
Always 1.5×	10.46%
Always 2.5×	10.15%
Always 3×	8.43%



Takeaway: constant leverage barely helps at 1.5× and *loses* by 3×. Across a full century — including 1929 and 2008 — naive "just hold 3×" underperforms plain buy & hold, and badly trails the trend rule. Something is eating the extra exposure. That something is volatility decay.

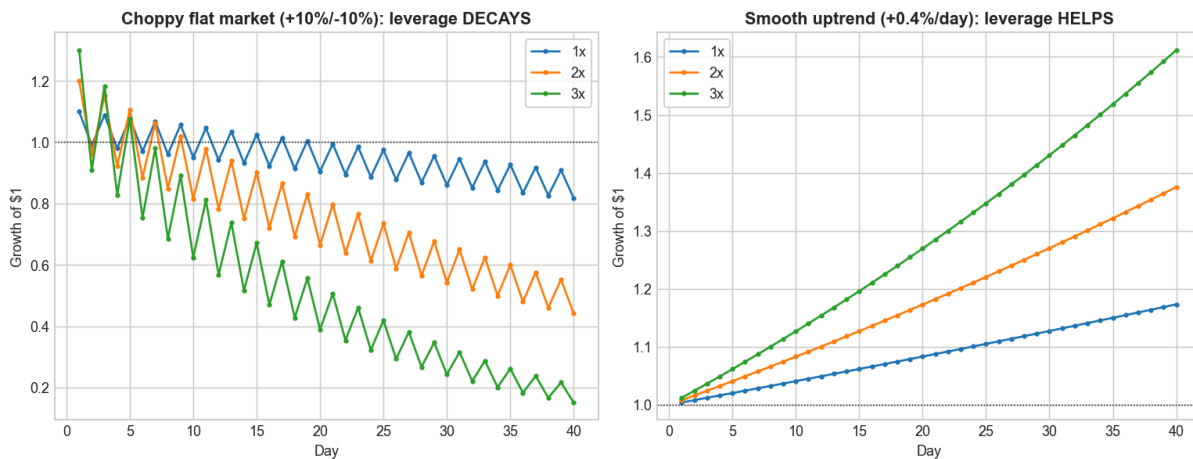
Step 3 — Volatility decay, and why *timing* matters

The arithmetic. A +10% day followed by a −10% day:

	1×	2×	3×
Two-day return	−1.0%	−4.0%	−9.0%

The market round-trips to roughly flat, but leverage loses — and the loss grows with the *square* of leverage. The annual penalty is the **variance drag** $\approx \frac{1}{2} \cdot L^2 \cdot \sigma^2$: at 20% volatility that is ~2%/yr for 1×, **8%/yr for 2×**, **18%/yr for 3×**.

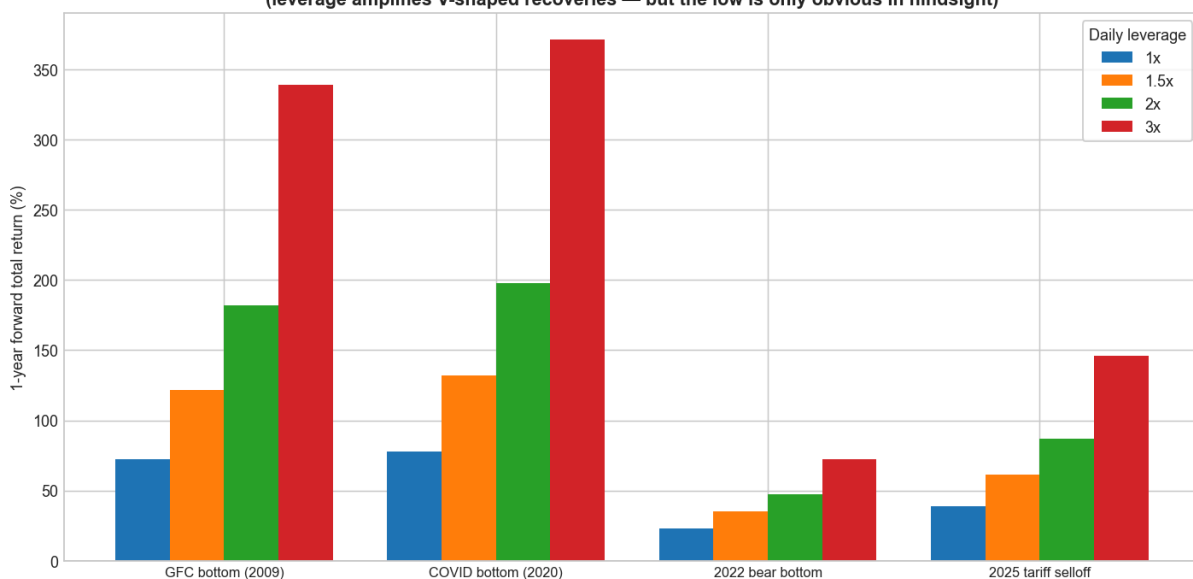
Volatility decay: the SAME daily leverage helps or hurts depending on the path



But decay is a property of choppy/falling markets. In a smooth uptrend, leverage amplifies gains. The sharpest example is buying leverage at a **market bottom**, where the rebound is steep and one-directional. Forward **1-year** total return if you had bought at the exact low:

Bottom	1×	1.5×	2×	3×
GFC (2009-03-09)	+72%	+122%	+182%	+339%
COVID (2020-03-23)	+78%	+132%	+198%	+372%
2022 (2022-10-12)	+23%	+35%	+47%	+72%
2025 tariff selloff (2025-04-08)	+39%	+61%	+87%	+146%

If you BUY leverage at the exact bottom: 1-year forward return
(leverage amplifies V-shaped recoveries — but the low is only obvious in hindsight)



Takeaway: leverage absolutely *can* work in your favour — **if you time it**, buying into the violent recovery off a low. The catch, which drives the rest of the paper: *the low is only obvious in hindsight*. The question is whether a simple, rules-based signal (the 200-day MA) can capture enough of that timing to make leverage pay — without knowing the bottom in advance.

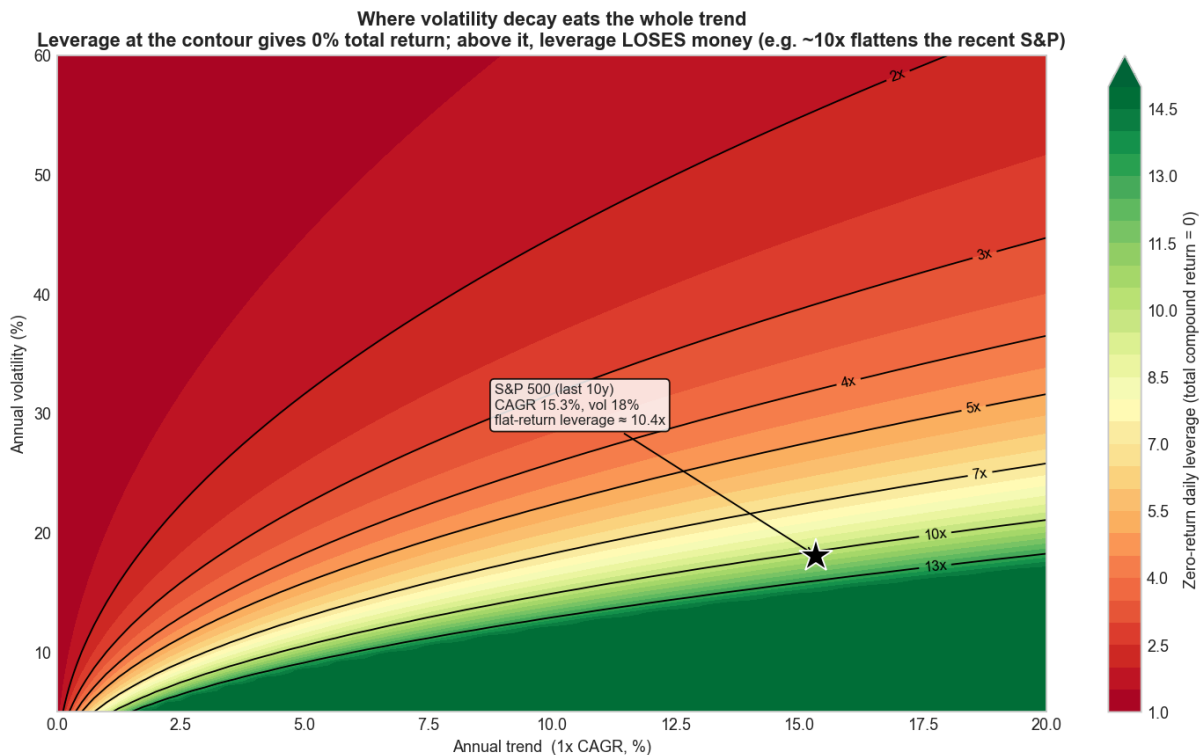
Step 4 — How much leverage is "too much"?

Two closed-form leverage thresholds, each a function of the trend (the $1\times$ CAGR g) and the annual volatility σ . Both are confirmed by Monte Carlo.

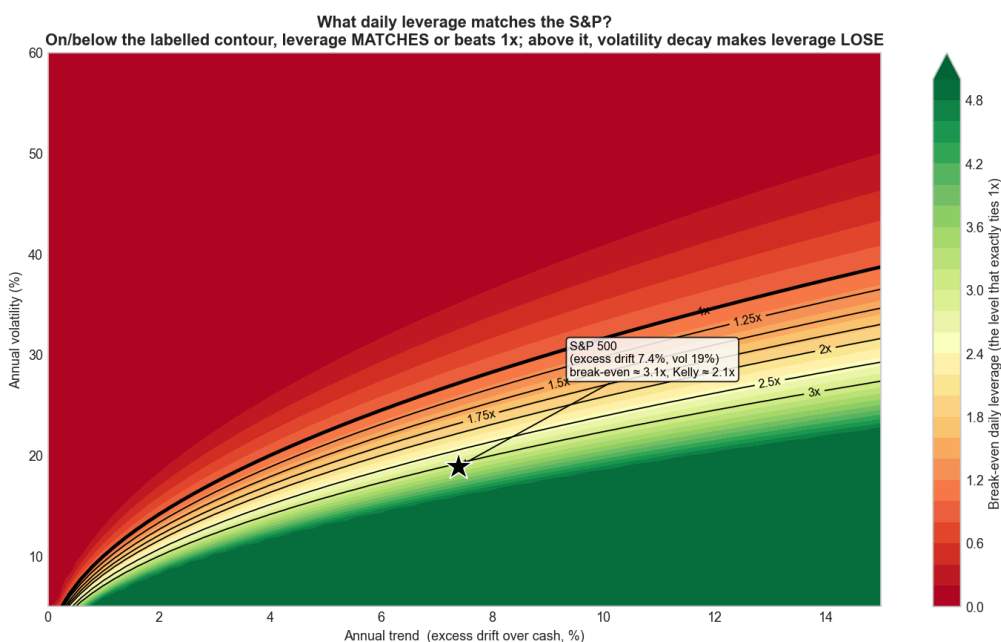
(a) The "flat total return" leverage. Leveraged compound growth is $g(L) = L \cdot \mu - \frac{1}{2} \cdot L^2 \cdot \sigma^2$. Setting it to zero gives the leverage at which volatility decay exactly eats the whole trend:

$$L_{\text{zero}} = \frac{2g}{\sigma^2} + 1$$

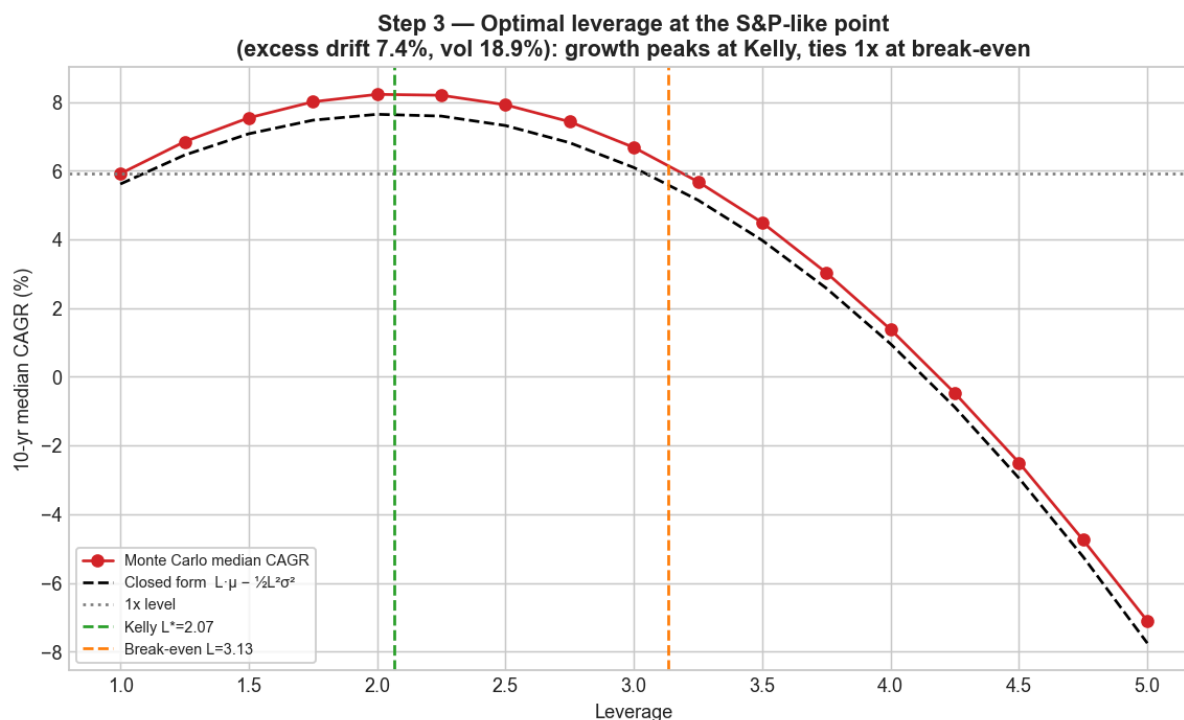
Below it, leverage still grows; above it, you **lose money** outright; far above it (e.g. $\sim 10\times$ +) you are effectively wiped out. For the S&P over the **last 10 years** (CAGR 15.3%, vol 18.1%), the flat-return leverage is $\approx 10.4\times$ — which is why a hypothetical "10x" S&P fund would have gone essentially nowhere despite a strong decade.



(b) The "matches 1x" leverage. The leverage whose compound return merely *ties* buy & hold is $L = 2\mu/\sigma^2 - 1$ (in excess-return terms). For the S&P (excess drift 7.4%, vol 18.9%) that is $\approx 3.1\times$, with the growth-optimal **Kelly** leverage $\mu/\sigma^2 \approx 2.1\times$ sitting exactly halfway between $1\times$ and the break-even.



A fine-grid Monte Carlo lands precisely on the closed form — median CAGR peaks at Kelly and returns to the 1× level at break-even:



Takeaway: for the S&P's drift and volatility, *steady* leverage beyond ~2× (Kelly) buys little extra growth, beyond ~3× you fall behind 1×, and around ~10× you make nothing. Leverage has a sweet spot that depends entirely on the trend-to-volatility ratio. Since the trend rule lets us choose *when* to be leveraged, the natural idea is to apply leverage only when conditions are favourable.

Step 5 — The switching strategy (200-day MA)

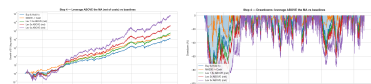
We now switch between **daily-leveraged** and **ordinary 1× S&P exposure** using the 200-day MA. There are two opposite ways to do it, and we test both.

Strategy A — "buy leverage low" (leverage BELOW the MA). When the market is below trend (cheap, "on sale"), hold $L \times$ leverage to ride the rebound; when above trend, hold plain 1×. This is the intuitive version motivated by the bought-the-low event studies in Step 3.

Strategy B — "the other way" (leverage ABOVE the MA). Leverage the calm, above-trend regime; drop to plain 1× when the market falls below trend.

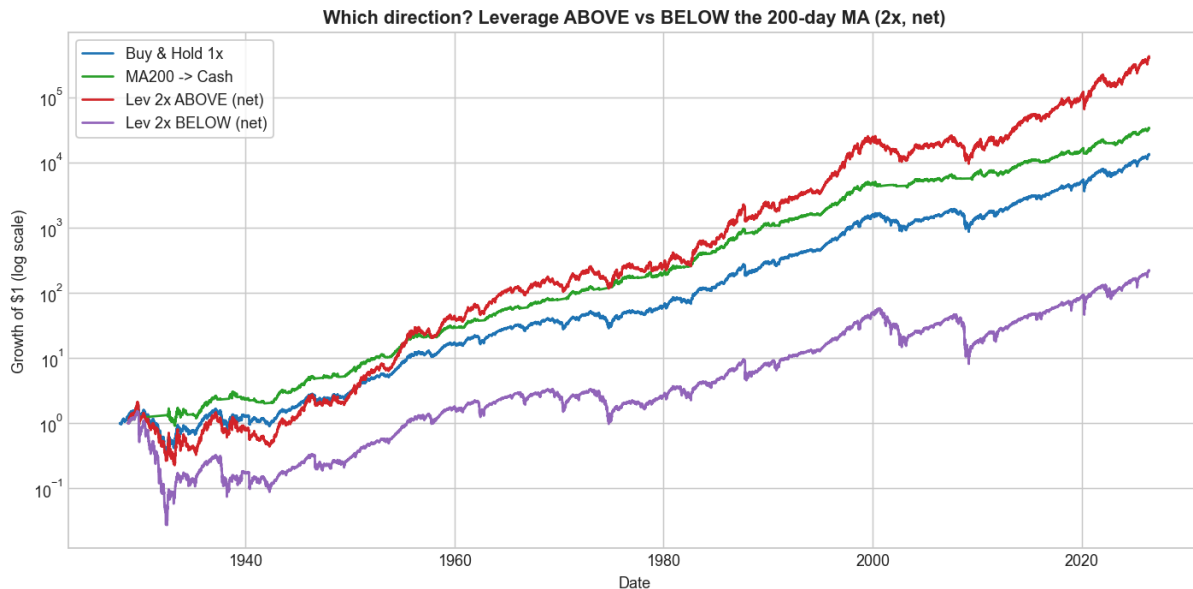
Full history (1928–2026), **net of costs**, both directions at every level:

Strategy	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar
Buy & Hold 1×	10.1%	18.9%	0.40	0.56	−83.9%	0.12
MA200 → Cash	11.3%	12.6%	0.60	0.84	−46.2%	0.24
A: Lev 1.5× BELOW	7.9%	24.7%	0.27	0.38	−94.3%	0.08
A: Lev 2× BELOW	5.7%	31.0%	0.21	0.29	−98.2%	0.06
A: Lev 3× BELOW	−0.2%	44.3%	0.13	0.19	−99.9%	−0.00
B: Lev 1.5× ABOVE	11.9%	23.6%	0.43	0.60	−85.7%	0.14
B: Lev 2× ABOVE	14.2%	28.9%	0.47	0.66	−89.2%	0.16
B: Lev 3× ABOVE	17.5%	40.4%	0.50	0.71	−95.7%	0.18



The intuitive idea fails. Strategy A ("buy leverage low") gets *worse* as leverage rises — 1.5× below trails buy & hold, 2× below earns just 5.7%, and 3× below actually *loses money* (−0.2% CAGR) with a −99.9% drawdown. Why? The 200-day MA does **not** buy the bottom. It flags "below trend" at the *start* of a decline, when the market is still highly volatile and often has much further to fall (1929, 2000–02, 2008). So Strategy A piles leverage into exactly the high-volatility, still-falling regime where decay is worst — the opposite of the clean bought-the-low trades in Step 3.

The other way is far better. Strategy B (leverage the uptrend) *improves* with leverage and beats buy & hold on CAGR, Sharpe, Sortino, and Calmar at every level. At 2×, the two directions of the *same idea* differ enormously — Sharpe 0.47 vs 0.21, CAGR 14.2% vs 5.7%:



A closer look — 2000–2026 (drops the un-survivable 1929/1987 single-day crashes; net of costs):

Strategy	CAGR	Sharpe	Max DD	Calmar
Buy & Hold 1×	8.3%	0.41	−55.3%	0.15
MA200 → Cash	7.5%	0.52	−20.6%	0.36
Lev 1.5× ABOVE	9.5%	0.43	−59.0%	0.16
Lev 2× ABOVE	11.1%	0.45	−62.8%	0.18
Lev 3× ABOVE	13.4%	0.47	−73.5%	0.18
Lev 2× BELOW (A)	5.7%	0.28	−85.9%	0.07

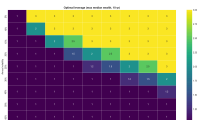
Post-2000 the ranking is identical and the above-MA drawdowns are far more survivable (3× peaks at −74% rather than −96%).

Two important caveats on Strategy B.

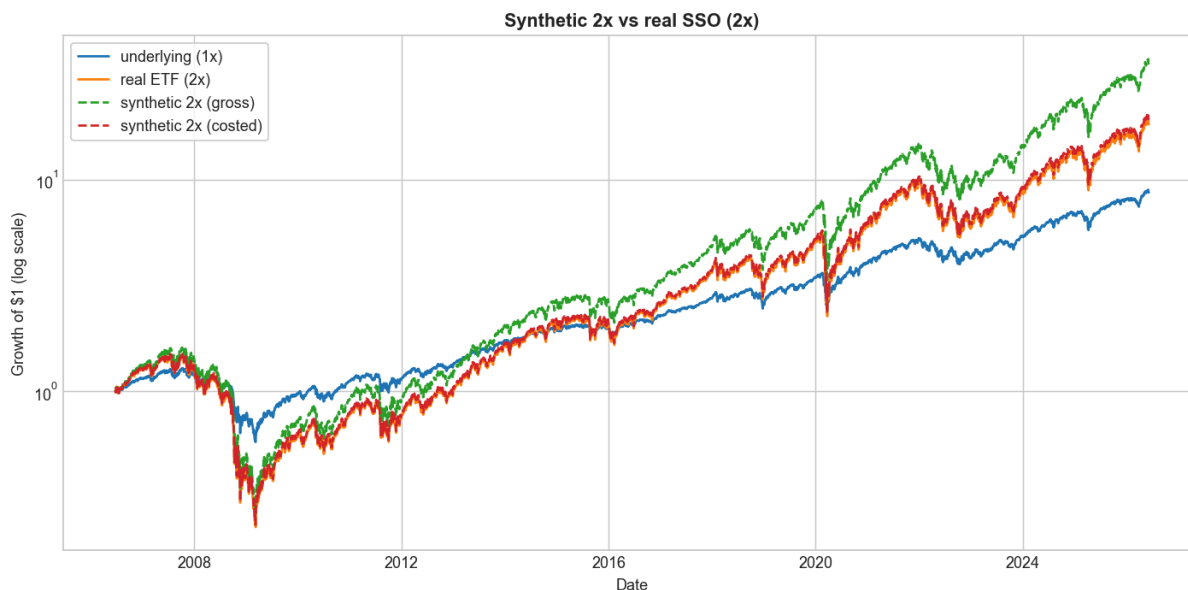
1. It beats buy & hold on every *ratio*, but it has **deeper maximum drawdowns** than buy & hold — you are leveraged *going into* fast crashes the MA can't dodge in time (a −22% day at 3× is −66% in one day).
2. It does **not** beat the plain move-to-cash rule on a risk-adjusted basis: MA200→cash still has the best Sharpe, Sortino, Calmar, and (by far) the shallowest drawdown.

Step 6 — Robustness & verdict

Monte Carlo (optimal leverage by regime). Simulating 10,000 paths across a grid of drifts and volatilities shows the same diagonal frontier: high leverage only pays in **low-volatility, positive-drift** regimes; at ≥30–40% volatility the optimum collapses to 1×. Below-trend markets are precisely the high-volatility rows — which is why Strategy A fails and Strategy B (leveraging the calm, above-trend rows) works.



Real leveraged ETFs. SSO (2×), UPRO/SPXL (3×) track their daily multiple closely — realized betas 1.95 / 2.97 / 2.93 — with 3.1–4.6%/yr tracking error, and our costed *synthetic* leverage slightly *overstates* the real ETFs (by 0.3–0.6%/yr). So the synthetic results are, if anything, a touch optimistic; real products would not have done better.



Costs matter for Strategy B. Because B is leveraged ~70% of the time, financing is a real drag: 3× above-MA falls from 26.8% CAGR gross to 17.5% net. All Strategy-B numbers above are net of expense + financing + turnover.

Verdict (provisional — headline to be finalised).

- The **200-day MA adds genuine value**: move-to-cash is the best *risk-adjusted* rule tested (Sharpe 0.60, drawdown –46%).
- **Leverage can work, but only with the right timing and direction.** Buying leverage at true lows is spectacular but unknowable in advance; the MA's "below-trend" signal is the *wrong* time to leverage (high volatility, still falling), so the intuitive "buy leverage low" rule fails.
- **Leveraging the *uptrend* (above the MA) is the productive direction** — it beats buy & hold on CAGR/Sharpe/Sortino/Calmar across ~100 years — but at the cost of deeper drawdowns, and it still does not beat simple move-to-cash on risk-adjusted terms.
- Practical reading: trend-following is fundamentally a *risk-reducer*; if you add leverage, add it **modestly (~1.5–2×) above the trend**, never below it.

Limitations

- True daily *total* return begins in 1988; pre-1988 daily history is a validated reconstruction (0.5%/yr tracking error), and the pre-1960 cash rate is a documented constant.
- Daily-leverage modelling assumes execution at the close with a fixed cost; real financing spreads, borrow availability, and slippage vary over time.
- Real leveraged ETFs are young (2006–2009) and born into a bull market.
- The Monte Carlo uses constant drift/volatility with i.i.d. (optionally fat-tailed) shocks; real volatility clustering punishes leverage *more*, so the i.i.d. "optimal leverage" is generous.
- US-only, single index, no taxes. One ~100-year path is still one sample.

Future work. Volatility targeting (scale leverage by `target_vol / realized_vol`) to tame Strategy B's deep drawdowns; multi-timeframe trend signals; international and longer histories; explicit taxes.

Reproducibility

```
pip install -r requirements.txt
python run_all.py                # broader sweep, Monte Carlo, ETF tests (supplementary)
python run_faber_leverage.py    # this paper: Faber replication, maps, event studies, switchi
python build_notebooks.py       # rebuilds notebooks/01..09
python -m pytest tests/ -q      # 11 sanity tests
python build_pdf.py             # optional: rebuild this PDF
```

Figures are PNGs in `charts/`; numbers are in `results/` (this paper's headline JSON is `results/headline_faber.json`; the supplementary sweep is `results/headline_results.json`). Data is cached in

`data/raw` (including the Shiller spreadsheet). Random seeds are fixed in `src/config.py`.

Educational research only — not investment advice.